

Navigate Advantage Access for Friis Epidemiology 101, Third Edition (Open Enrollment)
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CHAPTER 4 Data and Disease Occurrence
Introduction

Introduction

You probably have not given much thought to epidemiologic data. However, this is a cutting-edge topic that has extreme importance for our country and modern society. In the present chapter you will learn how high-quality data are linked to important functions such as performance of essential public health services, especially program and policy evaluation. A mastery of data-related skills can help you bring needed assets to public health and commercial venues.

This chapter extends the coverage of quantitative measures by providing information about sources of data that are used for epidemiologic research. Previously, you learned about epidemiologic measures derived from ratios, rates, proportions, and percentages. The epidemiologic measures that were described included counts and crude mortality rates as well as case fatality rates, proportional mortality ratios, specific mortality rates, and adjusted mortality rates. All of these measures are used to describe mortality in populations. In addition, the terms *incidence* and *prevalence* were defined and are used to describe morbidity. The present chapter will link these concepts with data sources.

Two vital concerns of epidemiology are, first, the quality of data available for describing the health of populations and, second, whether these data are being applied in an appropriate manner. Specifically, you will learn appropriate and inappropriate uses for particular types of data. For example, it is not feasible to calculate some epidemiologic measures without a qualified data source. This chapter will come in handy when you are seeking data for computing basic indices of morbidity and mortality. The information presented will also aid you in evaluating associations between exposures and health outcomes derived from epidemiologic research. Refer to **Table 4.1** for a list of important terms used in this chapter.

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TABLE 4.1 List of Important Terms Used in This Chapter

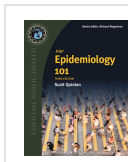
Data Acquisition Criteria for Data Quality

Big data	Appropriate uses of data
Data linkage	Availability of data
Data mining	Completeness of population coverage

Data Sources

American Community Survey
Morbidity surveys of the population
Public health surveillance

Data Acquisition	Criteria for Data Quality	Data Sources
MEDLINE	External validity	Registry data
Online retrieval	Nature of the data	Reportable disease statistics
Three Vs of big data	Personally identifiable information	U.S. Census Bureau
	Representativeness of data	Vital events



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CHAPTER 4 Data and Disease Occurrence

Epidemiology in the Era of Big Data

Epidemiology in the Era of Big Data

Increasingly, you have heard about big data. What exactly is meant by “**big data**?” This somewhat ambiguous term refers to vast electronic storehouses of information that include Internet search transactions, social media activities, data from health insurance programs, and electronic medical records from receipt of healthcare services, among many other sources. These data are relevant to epidemiology because they may cover entire populations or, at least, very large numbers of people. In addition, people can analyze big data to discover patterns of variables (distributions and determinants) associated with diseases and other health conditions. By combining big data, epidemiologists have new insights into the determinants of morbidity and mortality. However, the uses of big data have strengths as well as weaknesses.

Three qualities, known as the **three Vs**, characterize big data: volume, variety, and velocity.¹ Volume refers to the massive amount of data that requires flexible and expandable storage options. Variety refers to the many different formats of data, which may include free text, digital images, and social media sources. Velocity refers to the fact that many sources provide data in near real-time, which allows for rapid analysis and decision making.¹ More recently, two additional qualities of big data have been suggested: veracity, which refers to the quality, trustworthiness, and authenticity of the data;² and value, which refers to the many potential applications of big data.³

The vast troves of accumulated data include people’s social media accounts, online activities, and purchases in stores. It is technically possible to combine this information with health data, visits to doctors, hospital stays, and health insurance programs. The procedure known as **data linkage** is used to join data elements contained in databases by tying them together with a common identifier. Refer to **Figure 4.1** for an illustration of this process. Other data with potential for linkage include real-time transmissions from cellular telephones and the output from fitness tracking devices.

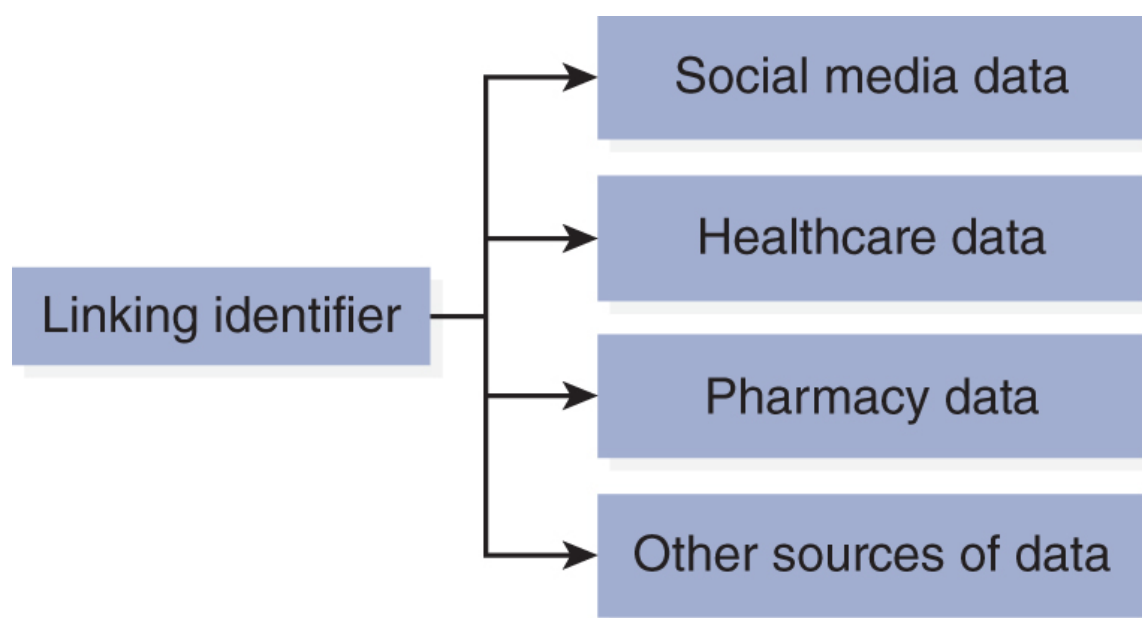


Figure 4.1 Epidemiology and big data

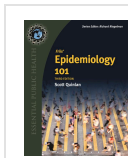
Some firms (often called data brokers) specialize in big data analytics and data mining. The process of **data mining** involves gathering and exploring large troves of data to discern heretofore unrecognized patterns and associations in the data. For example, data mining has transformed political campaigns by allowing campaigns to learn more about their constituents, such as which movies they stream, what shows they watch, and what products they buy. This has allowed these campaigns to be more targeted in their advertising.³

Google and Facebook track the activity of Internet users, as do online retailers. An illustration of applying the methodology of big data to health research is Google's introduction of Google Flu Trends (GFT) in the fall of 2008. The objective of GFT was to provide an early warning for influenza in advance of surveillance information from the Centers for Disease Control and Prevention (CDC).⁴ Subsequently, GFT was found to substantially overestimate the prevalence of influenza,⁵ thereby highlighting one of the limitations of using big data to predict disease occurrence. The application of big data was apparent during the COVID-19 pandemic as there was an increasing volume of available, digitized, global data that permitted data-driven decision making on such topics as universal masking, vaccination strategies, and social distancing.³

However, in addition to their promise of breakthroughs, the applications of big data are accompanied by caveats for epidemiologic research. One is the concern for personal privacy when the data are accessed in ways not originally intended by the individual who makes a purchase online or goes to the doctor. Another is that all of the rigorous criteria of science and epidemiologic research design need to be upheld. These include considerations regarding the reliability and validity of data, correct use of study designs, and recognition of the criteria of causal inference. Also, errors

such as misidentification of cases can occur when records are combined through record linkage. In addition, patterns of data points that have been linked by data mining may represent only relationships and not necessarily causal associations.

A growing area of interest is the role of artificial intelligence (AI) in conjunction with big data sources in medicine and public health. AI involves the use of computers to conduct tasks that usually require human intervention. AI systems can process large amounts of data of different types quickly and may allow big data sources to be used more effectively to improve health. These systems have already shown some success in processing medical image files to improve diagnostics. Future developments may allow AI systems to improve screening for different diseases, better model the spread of infectious diseases, and synthesize large amounts of research data into useful recommendations. As this area continues to advance, attention will also need to be paid to some of the limitations of AI, including ensuring ethical data use and avoiding the perpetuation of racial biases and inequities that may be lurking in the data.



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CHAPTER 4 Data and Disease Occurrence
Online Sources for Retrieval of Epidemiologic Information

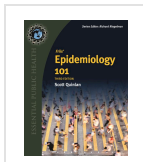
Online Sources for Retrieval of Epidemiologic Information

The process of **online retrieval** involves the acquisition of information from websites that are available to the public as well as from proprietary sites. Extensive resources are available for online retrieval of epidemiologic information, and the number of websites seems to be growing exponentially. Among the numerous websites that may be researched for data and other pertinent epidemiologic information are the following:

- Google (www.google.com): The Google site facilitates rapid access to epidemiologic documents and links. You can use Google most effectively when you create appropriate search terms. One may search for reports in written text format as well as for images and access to other materials.
- Centers for Disease Control and Prevention (CDC) (www.cdc.gov/): This site is the portal to many of the federal government's publications related to infectious and chronic diseases. One of these publications is the *Morbidity and Mortality Weekly Report*. The CDC Wonder site (<https://wonder.cdc.gov/>) is a repository of many different data sources related to population health, including vital statistics information.
- MEDLINE, National Library of Medicine (NLM), National Institutes of Health (NIH) (www.nlm.nih.gov/): MEDLINE is a site for performing bibliographic searches of health-related literature. You can search for citations to health-related literature on PubMed (www.ncbi.nlm.nih.gov/pubmed/).
- Websites of organizations and publications related to epidemiology:
 - The American Public Health Association (www.apha.org/): The American Public Health Association publishes the *American Journal of Public Health*.
 - The Society for Epidemiologic Research (www.epiresearch.org/), which sponsors the *American Journal of Epidemiology*.

- Professional organizations such as these sponsor regular conferences, which provide learning opportunities about epidemiology.

Navigate through the web and access these sites. Not only are they interesting in themselves, but also they will link you to many other related websites.



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CHAPTER 4 Data and Disease Occurrence
U.S. Census Bureau

U.S. Census Bureau

Measures of morbidity and mortality require accurate information about the size and characteristics of the population. The U.S. Census Bureau offers a plethora of data regarding the characteristics of our country. **Figure 4.2** portrays the bureau's logo with the date of Census 2020. One of the applications of census data is the clarification of denominators used in epidemiologic measures, such as rates and proportions. Also, many epidemiologic studies classify health outcomes according to sociodemographic variables; consequently, accurate information about these characteristics is needed.

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April 1:
Census day

Figure 4.2 U.S. Census Bureau

U.S. Department of Commerce. U.S. Census Bureau. Available at:
https://www.census.gov/content/dam/Census/library/visualizations/2016/comm/cb16-61_graphic.pdf. Accessed June 27, 2016.

Description

You can obtain data and related products from the U.S. Census Bureau by accessing the Census website (www.census.gov) and exploring different aspects of the Census data online (<https://data.census.gov/>). The census

provides a wealth of data that can be used to define the denominator in epidemiologic measures. These data include official estimates of the total population size and subdivisions of the population by geographic area. The **U.S. Census Bureau** conducts a census of the population every 10 years (the decennial census—e.g., 1980, 1990, 2000, 2010, 2020, 2030, and beyond).

To keep U.S. population estimates current with changes due to births, deaths, and migration, the Census Bureau provides annual estimates in noncensus years. The Bureau creates annual estimates by starting with the population base from the most recent full census, for example, the population size on April 1, 2020, and adds the number of births and the net number of migrants and then subtracts the number of deaths. This same procedure is followed over successive years and produces estimates with a high degree of accuracy.⁶ Census 2020 employed a short data collection form only. Detailed information collected previously by the long questionnaire is now part of the Census Bureau's American Community Survey, which is completed regularly.⁷ (Refer to **Exhibit 4.1** for additional information about the 2020 Census.)

Exhibit 4.1 About the 2020 Census

The 2020 Census represented the largest participation movement ever witnessed in our country. The 2020 Census was the first time that all households were given the opportunity to respond to the census online. The 2020 Census also had to contend with the COVID-19 pandemic, which slowed down data collection and led to slightly later release dates for different parts of the census data. National and state population totals from the 2020 Census were released in April 2021. Redistricting data, which include additional state, county, and local counts, were released starting in August 2021.

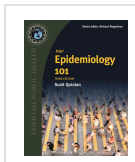
For the 2020 Census, the following information was collected for every person and housing unit in the United States:

- Age
- Hispanic or Latino origin
- Race
- Sex
- Whether the home is owned or rented
- Number of people living in the home
- Relationship of other people staying at the home

Since 1975, the Census Bureau has had the responsibility to produce small-area population data needed to redraw state legislative and congressional districts. Other important uses of Census data include the distribution of funds for government programs, such as Medicaid; planning the right locations for schools, roads, and other public facilities; helping real estate agents and potential residents learn about neighborhoods; and identifying trends over time that can help predict

future needs. Most census data are available for many levels of geography, including states, counties, cities and towns, ZIP Code Tabulation Areas, census tracts, blocks, and more.

Data from United States Census Bureau. Available at:
<https://www.census.gov/newsroom/blogs/random-samplings/2021/04/2020-census-data-review.html>. Accessed February 20, 2023.



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CHAPTER 4 Data and Disease Occurrence
The Vital Registration System and Vital Events

The Vital Registration System and Vital Events

Vital events are deaths, births, marriages, divorces, and fetal deaths. The vital registration system in the United States collects information routinely on these events. The legal authority for the registration of vital events within the United States is held by individual states, five U.S. territories (e.g., Puerto Rico), the Commonwealth of the Northern Mariana Islands, New York City, and Washington, D.C. These jurisdictions are charged with keeping records of vital events and providing certificates of marriage and divorce as well as birth and death certificates. In many instances, certificates that document vital events are also available from local health departments in the United States.

Deaths

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Data are collected routinely on all deaths that occur in the United States. Mortality data have the advantage of being almost totally complete because deaths are unlikely to go unrecorded in the United States. In many instances, the funeral director initiates completion of the death certificate and the attending physician then completes the section on date and cause of death. If the death occurred as the result of unintentional injury, suicide, or homicide, or if the attending physician is unavailable, then the medical examiner or coroner completes and signs the death certificate. Finally, the local registrar checks the certificate for completeness and accuracy and sends a copy to the state registrar. The state registrar also checks for completeness and accuracy and sends a copy to the National Center for Health Statistics (NCHS), which compiles and publishes national mortality rates (e.g., in *National Vital Statistics Reports*). Each state has its own system for investigating deaths that occur. Some states use medical examiner-based systems, while others use coroner-based systems, and some use a combination of the two (refer to **Figure 4.3**). These differences are important to note as they may impact the accuracy of the information included on the death certificate. In most cases, coroners are not required to have any medical training and are often elected or appointed, while medical examiners are required to have medical training.⁸

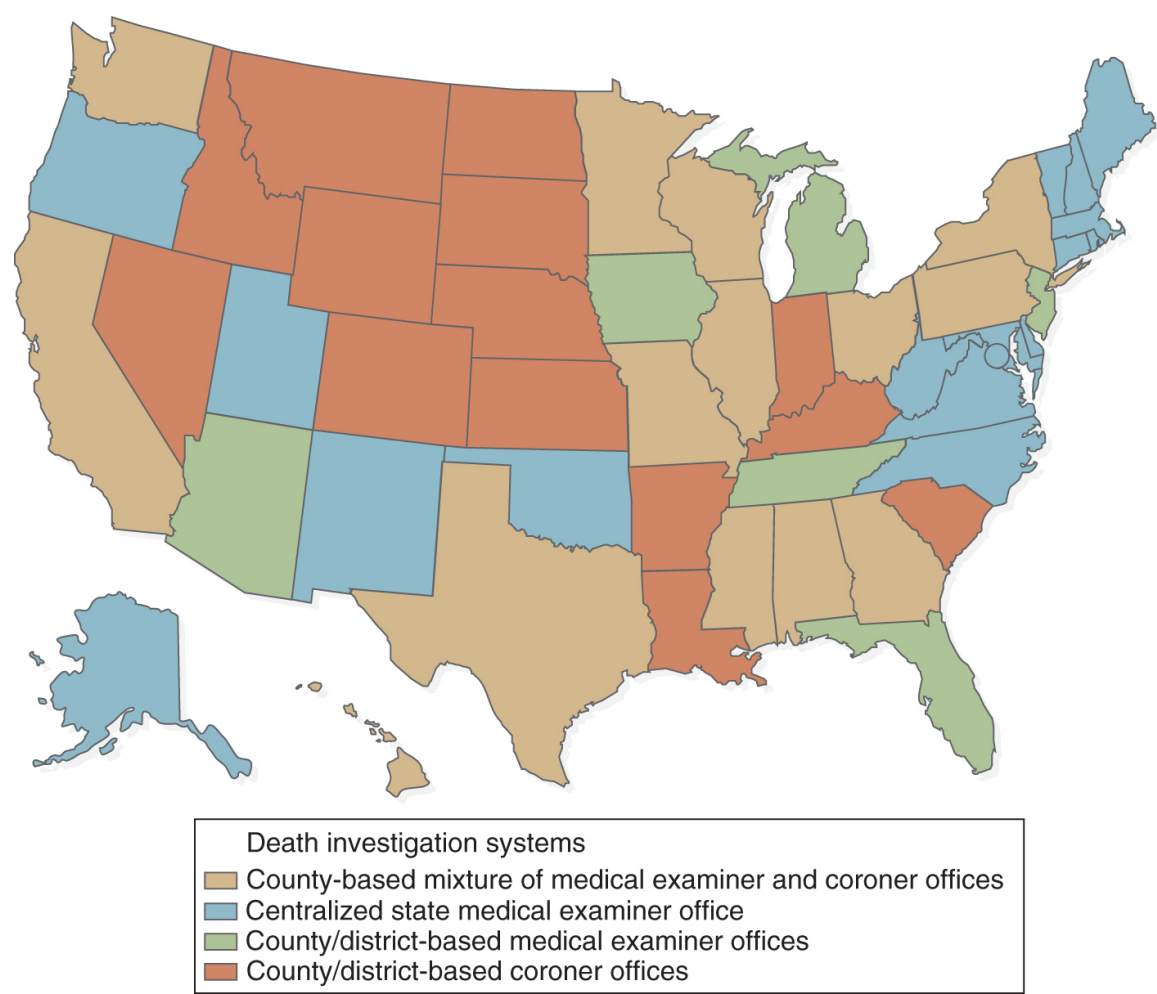


Figure 4.3 Death investigation systems by state, United States

Centers for Disease Control and Prevention. Death investigations. Available at: <https://www.cdc.gov/phlp/publications/coroner/death.html>. Accessed February 20, 2023.

Description

Death certificate data in the United States include information about the decedent shown in **Table 4.2**. An example of a death certificate and additional information regarding the kinds of data collected are shown in **Figure 4.4**.

TABLE 4.2 Typical Data Recorded in Death Certificates

Demographic characteristics	Date and place of death—hospital or elsewhere
- Age - Sex - Race	- Cause of death - Immediate cause - Contributing factors

U.S. STANDARD CERTIFICATE OF DEATH									
LOCAL FILE NO.					STATE FILE NO.				
1. DECEDENT'S LEGAL NAME (Include AKAs if any) (First, Middle, Last)					2. SEX		3. SOCIAL SECURITY NUMBER		
4a. AGE-Last Birthday (Years)		4b. UNDER 1 YEAR Months Days		4c. UNDER 1 DAY Hours Minutes		5. DATE OF BIRTH (Mo/DatYr)		6. BIRTHPLACE (City and State or Foreign Country)	
7a. RESIDENCE-STATE			7b. COUNTY			7c. CITY OR TOWN			
7d. STREET AND NUMBER				7e. APT. NO.		7f. ZIP CODE		7g. INSIDE CITY LIMITS? ☐ Yes ☐ No	
8. EVER IN US ARMED FORCES? ☐ Yes ☐ No		9. MARITAL STATUS AT TIME OF DEATH ☐ Married ☐ Married, but separated ☐ Widowed ☐ Divorced ☐ Never Married ☐ Unknown			10. SURVIVING SPOUSE'S NAME (If wife, give name prior to first marriage)				
11. FATHER'S NAME (First, Middle, Last)					12. MOTHER'S NAME PRIOR TO FIRST MARRIAGE (First, Middle, Last)				
13a. INFORMANT'S NAME			13b. RELATIONSHIP TO DECEDENT			13c. MAILING ADDRESS (Street and Number, City, State, Zip Code)			
14. PLACE OF DEATH (Check only one: see instructions)									
IF DEATH OCCURRED IN A HOSPITAL: ☐ Inpatient ☐ Emergency Room/Outpatient ☐ Dead on Arrival					IF DEATH OCCURRED SOMEWHERE OTHER THAN A HOSPITAL: ☐ Hospice facility ☐ Nursing home/Long term care facility ☐ Decedent's home ☐ Other (Specify):				
15. FACILITY NAME (If not institution, give street & number)					16. CITY OR TOWN, STATE, AND ZIP CODE			17. COUNTY OF DEATH	
18. METHOD OF DISPOSITION: ☐ Burial ☐ Cremation ☐ Donation ☐ Entombment ☐ Removal from State ☐ Other (Specify):					19. PLACE OF DISPOSITION (Name of Cemetery, Crematory, other place)				
20. LOCATION-CITY, TOWN, AND STATE					21. NAME AND COMPLETE ADDRESS OF FUNERAL FACILITY				
22. SIGNATURE OF FUNERAL SERVICE LICENSEE OR OTHER AGENT							23. LICENSE NUMBER (Of Licensee)		
ITEMS 24-28 MUST BE COMPLETED BY PERSON WHO PRONOUNCED OR CERTIFIES DEATH					24. DATE PRONOUNCED DEAD (Mo/DatYr)		25. TIME PRONOUNCED DEAD		
26. SIGNATURE OF PERSON PRONOUNCING DEATH (ONLY When applicable)					27. LICENSE NUMBER		28. DATE SIGNED (Mo/DatYr)		
29. ACTUAL OR PRESUMED DATE OF DEATH (Mo/DatYr) (Spell Month)			30. ACTUAL OR PRESUMED TIME OF DEATH			31. WAS MEDICAL EXAMINER OR CORONER CONTACTED? ☐ Yes ☐ No			
CAUSE OF DEATH (See instructions and examples) 32. PART I. Enter the chain of events—diseases, injuries, or complications—that directly caused the death. Do NOT enter terminal events such as cardiac arrest, respiratory arrest, or ventricular fibrillation without showing the etiology. DO NOT ABBREVIATE. Enter only one cause on a line. Add additional lines if necessary. IMMEDIATE CAUSE (Final disease or condition—resulting in death) a. _____ Due to (or as a consequence of): _____ Sequentially list conditions, if any, leading to the cause listed on line a. Enter the UNDERLYING CAUSE (disease or injury that initiated the events resulting in death) LAST b. _____ Due to (or as a consequence of): _____ c. _____ Due to (or as a consequence of): _____ d. _____									Approximate interval: Onset to death _____ _____ _____

Figure 4.4 U.S. standard certificate of death*

*Revised November 2003.

Centers for Disease Control and Prevention. Available at:

<http://www.cdc.gov/nchs/data/dvs/DEATH11-03final-acc.pdf>. Accessed February 20, 2023.

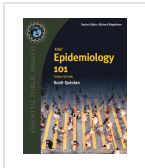
Description

Mortality data are one of the most commonly used indices in public health. Although available and fairly complete, death certificates may not be entirely accurate regarding the assigned cause of death. When an older person with a chronic illness dies, the primary cause of death may be unclear. Death certificates list multiple causes of mortality as well as the underlying cause. However, assignment of the cause of death sometimes may be arbitrary. In illustration, diabetes may not be given as the immediate cause of death; rather, the certificate may list the cause of death as heart failure or pneumonia, each of which could be a complication of diabetes. Another factor that detracts from the accuracy of death certificates is lack of standardization of diagnostic criteria employed by various physicians in different hospitals and settings. Yet another problem is the stigma associated with certain diseases. For example, if the decedent died as a result of human immunodeficiency virus (HIV) infection or alcoholism and was a long-time friend of the attending physician, the physician may be

reluctant to specify this information on a document that is available to the general public. Death certificate data received renewed attention during the COVID-19 pandemic as deaths from COVID-19 were counted and reported. In many cases, the immediate cause of death was not COVID-19, but instead the sequelae of COVID-19 such as pneumonia, respiratory failure, or the worsening of existing health conditions.

Birth Statistics

Birth statistics include live births and fetal deaths. Presumably, birth and fetal death statistics are nearly complete in their coverage of the general population. (Refer to **Table 4.3** for a list of information collected by certificates of live birth and reports of fetal death.) One of the uses of birth certificate data is for calculation of birth rates. Information is also collected about a range of conditions that may affect the neonate, including conditions present during pregnancy, congenital malformations, obstetric procedures, birth weight, length of gestation, and demographic background of the mother. Some of the data may be unreliable, reflecting possible inconsistencies and gaps in the mother's recall of events during pregnancy. Still another concern is that some malformations and illnesses may not be detected at the time of birth if, for example, the newborn appears to be normal. Many of the foregoing deficiencies of birth certificates also apply to the data contained in certificates of fetal death. In addition, state-to-state variations in requirements for fetal death certificates further reduce their utility for epidemiologic studies. Nevertheless, birth and fetal death certificate data have been used in many types of epidemiologic research. One of these is research concerning environmental influences on congenital malformations. For example, these data have been used to search for clusters of birth defects in geographic areas where mothers may have been

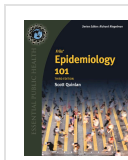


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The Vital Registration System and Vital Events

exposed to possible teratogens (agents that cause fetal malformation), such as pesticides or toxic pollutants.

TABLE 4.3 Examples of Information Collected by Birth and Fetal Death Certificates

Variable	U.S. Certificate of Live Birth	U.S. Report of Fetal Death
Name	Child's name	Name of fetus (optional)
Disposition (e.g., burial)	Not applicable	Disposition of fetus
Location	Facility name	Where delivered
Mother identifying information	Name, age, and place of residence	Name, age, and place of residence
Mother	Height, weight, number of previous live births	Height, weight, number of previous live births
Conditions contributing to fetal death	Not applicable	Initiating cause and other causes (e.g., pregnancy complications, fetal anomalies)
Risk factors in pregnancy	Diabetes, hypertension, infections	Diabetes, hypertension, infections
Congenital anomalies	Anencephaly, cleft palate, Down syndrome	Anencephaly, cleft palate, Down syndrome
Father	Name and age	Name and age



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CHAPTER 4 Data and Disease Occurrence
Data from Public Health Surveillance Programs

Data from Public Health Surveillance Programs

Public health surveillance programs are used for communicable and infectious diseases, noninfectious diseases, and risk factors for disease. **Public health surveillance** refers to the systematic and continuous gathering of information about the occurrence of diseases and other health phenomena. As part of the surveillance process, personnel analyze and interpret the data they have collected and distribute the data and associated findings to planners, health workers, and members of the public health community.

The public health community has been concerned with the possibility of using surveillance systems for detecting diseases associated with bioterrorism as well as early detection of disease outbreaks in general. **Figure 4.5** shows a worker protected against biologic disease agents. The term *syndromic surveillance* describes a procedure "...to identify illness clusters early, before diagnoses are confirmed and reported to public health agencies, and to mobilize a rapid response, thereby reducing morbidity and mortality."⁹ Epidemiologists think of syndromic surveillance as an early warning system for disease outbreaks.



Figure 4.5 A CDC laboratorian at work in a maximum containment, or “hot lab”

Centers for Disease Control and Prevention. Public Health Image Library, ID# 5538. Available at: <http://phil.cdc.gov/phil/home.asp>. Accessed March 8, 2016.

During the COVID-19 pandemic, the CDC launched the National Wastewater Surveillance System (NWSS) in September 2020. The system detects SARS-CoV-2 levels in community wastewater to provide an early warning that COVID-19 is spreading in the community. Wastewater, sometimes referred to as sewage, includes water from household or building use (such as toilets, showers, and sinks) that can contain human fecal waste, as well as water from nonhousehold sources (such as rain and industrial use). Health departments can then use these surveillance data to act quickly to prevent further spread of infection. Wastewater surveillance can be particularly useful in communities that do not have easy access to health care or where clinical testing is underused or unavailable. This form of surveillance has gained recent attention for tracking COVID-19 but has been used for some time to track other infections, namely poliovirus.¹⁰ For example, wastewater surveillance was used in New York State to detect poliovirus as an early warning of potential infections in late 2022.¹¹

Surveillance programs operate at the local, national, and international level. Here are some additional examples of surveillance systems:

- Communicable and infectious diseases. In the United States, healthcare providers and related workers send reports of diseases (known as notifiable and reportable diseases) to local health departments, which in turn forward them to state health departments and then to the CDC.

The CDC reports the occurrence of internationally quarantinable diseases (e.g., plague, cholera, and yellow fever) to the World Health Organization.

- Noninfectious diseases. Surveillance programs often focus on the collection of information related to chronic diseases, such as asthma.
- Risk factors for chronic diseases. The Behavioral Risk Factor Surveillance System (BRFSS) was established by the CDC to collect information on behavior-related risk factors for chronic disease. One of the tasks of the BRFSS is the monitoring of health-related quality of life in the United States.

Figure 4.6 gives an overview of a simplified surveillance system, which shows how reports of cases of disease (e.g., infectious diseases) move up the hierarchy. Potential reporting sources are physicians and other healthcare providers as well as workers in clinical laboratories and other health-related facilities. Data recipients include county health departments at the primary level, state health departments at the secondary level, and federal agencies at the tertiary level. Data recipients at all of these levels are involved in feedback and dissemination of information required for appropriate public health action. **Exhibit 4.2** and the section on reportable disease statistics describe these activities in more detail.

Exhibit 4.2 National Notifiable Diseases Surveillance System History

In 1878, Congress authorized the U.S. Marine Hospital Service (the forerunner of the Public Health Service [PHS]) to collect morbidity reports regarding cholera, smallpox, plague, and yellow fever from U.S. consuls overseas. This information was to be used for instituting quarantine measures to prevent the introduction and spread of these diseases into the United States. In 1879, a specific Congressional appropriation was made for the collection and publication of reports of these notifiable diseases. The authority for weekly reporting and publication of these reports was expanded by Congress in 1893 to include data from states and municipal authorities. To increase the uniformity of the data, Congress enacted a law in 1902 directing the Surgeon General to provide forms for the collection and compilation of data and for the publication of reports at the national level. In 1912, state and territorial health authorities—in conjunction with PHS—recommended immediate telegraphic reporting of five infectious diseases and the monthly reporting, by letter, of 10 additional diseases. The first annual summary of The Notifiable Diseases in 1912 included reports of 10 diseases from 19 states, the District of Columbia, and Hawaii. By 1928, all states, the District of Columbia, Hawaii, and Puerto Rico were participating in national reporting of 29 specified diseases. At their annual meeting in 1950, the State and Territorial Health Officers authorized a conference of state and territorial epidemiologists whose purpose was to determine which diseases should be reported to PHS. In 1961, CDC assumed responsibility for the collection and publication of data concerning nationally notifiable diseases.

The list of nationally notifiable infectious diseases and conditions is revised periodically. An infectious disease or condition might be added to the list as a new pathogen emerges, or a disease or condition might be removed as its incidence declines. Public health officials at state and territorial health departments collaborate with CDC staff in determining which infectious diseases and conditions should be considered nationally notifiable. The Council of State and Territorial Epidemiologists (CSTE), with input from CDC, makes recommendations annually for additions and deletions to the list. The list of infectious diseases and conditions considered reportable in each jurisdiction varies over time and across jurisdictions.

Data on selected notifiable infectious diseases are published weekly in the *Morbidity and Mortality Weekly Report (MMWR)*. Annual summaries of notifiable diseases are published online through the CDC Wonder data repository.

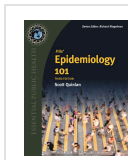
Data from Centers for Disease Control and Prevention. National Notifiable Diseases Surveillance System. History and background. Available at: <https://www.cdc.gov/nndss/index.html>. Accessed

February 20, 2023; Centers for Disease Control and Prevention. Nationally notifiable infectious diseases and conditions, United States: annual tables. Available at: https://wonder.cdc.gov/nndss/nndss_annual_tables_menu.asp. Accessed February 20, 2023.

Reportable and Notifiable Disease Statistics

By legal statute, physicians and other healthcare providers must report cases of certain diseases, known as reportable and notifiable diseases, to health authorities. **Reportable disease statistics** are statistics derived from diseases that physicians and other healthcare providers must report to government agencies according to legal statute. The determination of what diseases are considered reportable is made at the state level. Reportable diseases are usually infectious and communicable diseases that might endanger a population. Examples include sexually transmitted diseases, rubella, tetanus, measles, plague, and foodborne disease. In addition to these diseases, individual states may elect to maintain reports of communicable and noncommunicable diseases of local concern. Each year the CDC specifies diseases that are considered notifiable and requests that states submit reports of such diseases.¹² To supplement the notifiable disease surveillance system, the CDC operates a surveillance system for several noteworthy diseases such as salmonellosis, shigellosis, and influenza. For example, reports of influenza are tracked from October through May.

Examples of nationally notifiable infectious diseases are shown in **Table 4.4**. Many of the notifiable diseases and conditions are reportable in all states, while others are reportable in some states. The list changes every so often. For example, 2019 novel



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coronavirus disease (COVID-19) was added to the list of nationally notifiable diseases in 2020 and mpox (formerly known as monkeypox) was added to the list of nationally notifiable diseases in 2022.

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TABLE 4.4 Examples of Nationally Notifiable Conditions—United States, 2022

- Anthrax
- COVID-19
- Botulism
- Gonorrhea
- Hepatitis A, hepatitis B, hepatitis C
- Human immunodeficiency virus (HIV) infection (acquired immune deficiency syndrome [AIDS] has been reclassified as HIV Stage III)
- Meningococcal disease
- Mumps
- Syphilis
- Tuberculosis
- Viral hemorrhagic fever (includes Ebola virus)
- Zika virus disease

Centers for Disease Control and Prevention. National Notifiable Diseases Surveillance System (NNDSS). 2022 Nationally Notifiable Conditions. Available at: <https://ndc.services.cdc.gov/event-codes-other-surveillance-resources/>. Accessed February 20, 2023.

The major deficiency of reportable and notifiable data for epidemiologic research purposes is the possible incompleteness of population coverage. First, not every person who develops a disease that is on this list of notifiable conditions seeks medical attention, as persons who are afflicted with asymptomatic and subclinical illnesses are unlikely to visit a physician. For example, an active case of typhoid fever will go unreported if the affected individual is unaware that he or she has the disease. Typhoid Mary illustrated this phenomenon. Another factor associated with lack of complete population coverage is the occasional failure of overwhelmed healthcare providers to fill out the required reporting forms. This shortcoming can occur if responsible individuals do not keep current with respect to the frequently changing requirements for disease reporting in a local area. Also, as discussed earlier, a physician may be unwilling to risk compromising the confidentiality of the physician–patient relationship, especially as a result of concern and controversy about reporting cases of diseases that carry social

stigma. For example, incompleteness of HIV reporting may stem from the potential sensitivity of the diagnosis. More severe and unusual diseases, such as diphtheria, are almost always reported. Referring to Table 4.4, you will see that anthrax is an example of a nationally notifiable disease. **Figure 4.7** provides an illustration of a person who has contracted anthrax. (Typhoid Mary will be discussed in the chapter on infectious diseases.)



Figure 4.7 Patient with anthrax, one of the nationally notifiable diseases

Centers for Disease Control and Prevention. Public Health Image Library #19826. Available at <https://phil.cdc.gov/Details.aspx?pid=19826>. Accessed June 27, 2016.

Chronic Disease Surveillance: The Example of Asthma

Asthma, a highly prevalent disease that incurs substantial medical and economic costs, is associated with inflammatory lung and airway conditions that can result in breathing difficulty, coughing, chest tightness, and other pulmonary symptoms. Severe asthma symptoms can be life threatening. Asthma surveillance programs provide data necessary for the development and evaluation of healthcare services for afflicted persons. Asthma surveillance data include data collected at both the state and national levels. National data are available on asthma prevalence, activity limitation, days of work or school lost, rescue and control medication use, asthma self-management education, physician visits, emergency department visits, hospitalizations due to asthma, and deaths due to asthma from National

Center for Health Statistics (NCHS) surveys and the Vital Statistics System. One of the key sources of state-level asthma surveillance data is the Behavioral Risk Factor Surveillance System, which will be discussed later in this chapter.¹³ A key measure used in asthma surveillance is the prevalence of asthma attacks (refer to **Figure 4.8**). Asthma attacks tend to be more common in females than in males and in people younger than 18 years of age than in adults.

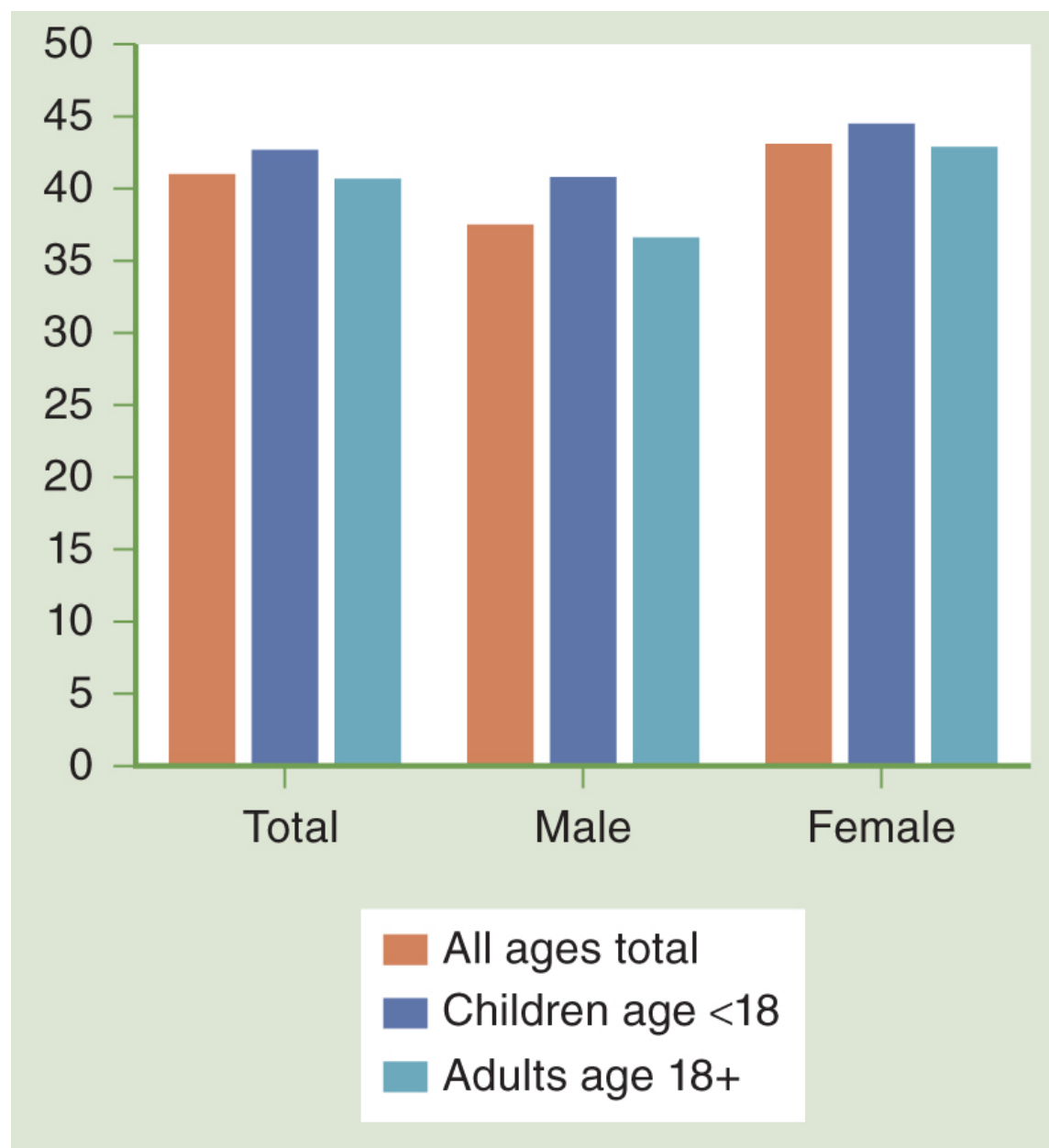


Figure 4.8 The prevalence of asthma attacks among those with current asthma by age and sex, 2020

Data from National Health Interview Survey, National Center for Health Statistics, CDC. Available at: <https://www.cdc.gov/asthma/nhis/2020/table6-1.htm>. Accessed on February 20, 2023.

Description

Behavioral Risk Factor Surveillance System

The Behavioral Risk Factor Surveillance System (BRFSS) is a noteworthy program used by the United States to monitor at the state level behavioral risk factors that are associated with chronic diseases. (See **Figure 4.9** for the logo of the BRFSS and **Exhibit 4.3** for a description of the program.) Public health experts regard the BRFSS as one of America's leading sources of behavioral risk factors, for example, smoking and lifestyle. It is also the world's largest ongoing health survey.

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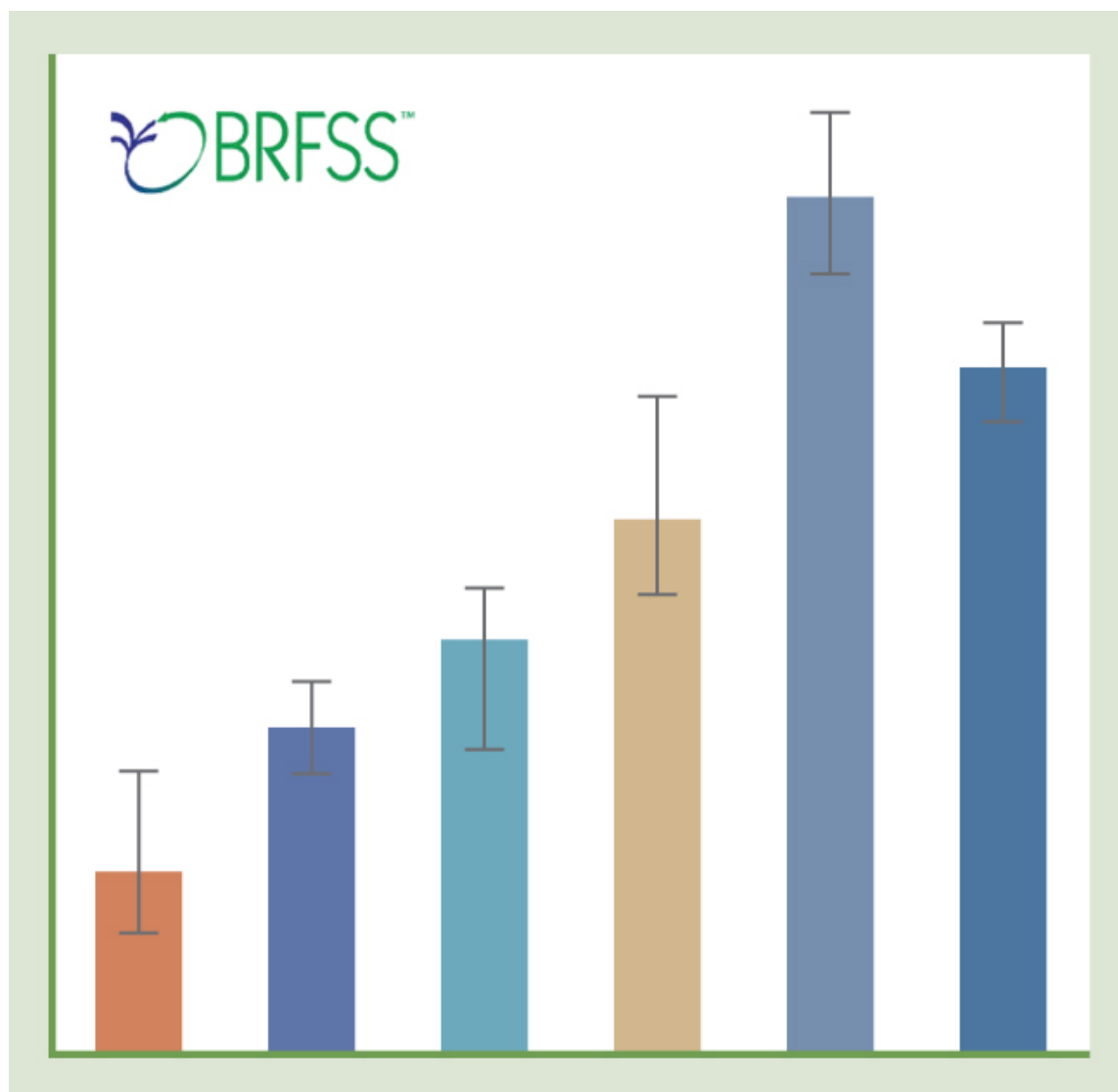


Figure 4.9 Behavioral Risk Factor Surveillance System (BRFSS)

Centers for Disease Control and Prevention. BRFSS prevalence & trends data. Available at: <http://www.cdc.gov/brfss/brfssprevalence/>. Accessed June 27, 2016.

Exhibit 4.3 Behavioral Risk Factor Surveillance System

The Behavioral Risk Factor Surveillance System (BRFSS) is the nation's premier system of health-related telephone surveys that collect state data about U.S. residents regarding their health-related risk behaviors, chronic health conditions, and use of preventive services. Established in 1984 with 15 states, BRFSS now collects data in all 50 states as well as the District of Columbia and three U.S. territories. BRFSS completes more than 400,000 adult interviews each year, making it the largest continuously conducted health survey system in the world. By collecting behavioral health risk data at the state and local level, BRFSS has become a powerful tool for targeting and building health promotion activities.

A Brief History

By the early 1980s, scientific research clearly showed that personal health behaviors played a major role in premature morbidity and mortality. Although national estimates of health risk behaviors among U.S. adult populations had been periodically obtained through surveys conducted by the National Center for Health Statistics (NCHS), these data were not available on a state-specific basis. This deficiency was viewed as a critical obstacle to state health agencies trying to target resources to reduce behavioral risks and their consequent illnesses. National data may not be applicable to the conditions found in any given state; however, achieving national health goals required state and local agency participation.

About the same time that personal health behaviors received wider recognition in relation to chronic disease morbidity and mortality, telephone surveys emerged as an acceptable method for determining the prevalence of many health risk behaviors among populations. In addition to their cost advantages, telephone surveys were especially desirable at the state and local level, where the necessary expertise and resources for conducting area probability sampling for in-person household interviews were not likely to be available.

As a result, surveys were developed and conducted to monitor state-level prevalence of the major behavioral risks among adults associated with premature morbidity and mortality. The basic philosophy was to collect data on actual behaviors, rather than on attitudes or knowledge, that would be especially useful for planning, initiating, supporting, and evaluating health promotion and disease prevention programs.

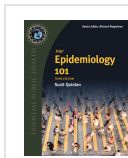
To determine feasibility of behavioral surveillance, initial point-in-time state surveys were conducted in 29 states from 1981–1983. In 1984, the CDC established the BRFSS, and 15 states participated in monthly data collection. Although the BRFSS was designed to collect state-level data, a number of states from the outset stratified their samples to allow them to estimate prevalence for regions within their respective states. CDC developed a standard core questionnaire for states to use to provide data that could be compared across states. Initial topics included smoking, alcohol use, physical inactivity, diet, hypertension, and seat belt use. Optional modules—standardized sets of questions on specific topics—were implemented in 1988.

BRFSS became a nationwide surveillance system in 1993. The questionnaire was redesigned to include rotating fixed core and rotating core questions and up to five emerging core questions. Approximately 100,000 interviews were completed in 1993.

BRFSS will mark its 40th year in 2023 and remains the gold standard of behavioral surveillance. Public health surveillance in the future will be much more complex and involve multiple ways of collecting public health data. Although telephone surveys will likely remain the mainstay of how BRFSS data are collected, it is likely that additional modes of interviewing will also be necessary. Since 2011, BRFSS has been conducting both landline telephone- and cellular telephone-based surveys. In 2014, all adults contacted through their cell phones were eligible to complete the survey, regardless of their landline phone use. By including cell phones in the survey, BRFSS is able to reach segments of the population that were previously inaccessible—those who have a cell phone but not a landline—and produce a more representative sample and higher quality data. To prepare for the future, BRFSS currently has several pilot studies and research initiatives underway. These efforts are critical for improving the quality of BRFSS data, reaching populations previously not included in the survey, and expanding the utility of the surveillance data.

Centers for Disease Control and Prevention, Behavioral Risk Factor Surveillance System. About BRFSS. Available at: <http://www.cdc.gov/brfss/about/> and http://www.cdc.gov/brfss/about/about_brfss.htm. Accessed February 20, 2023.

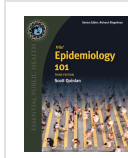
However, because the BRFSS is operated at the state level, the data may not be adequate for analyses at finer levels of aggregation such as counties. Moreover, sufficient information may not be available regarding health topics not specifically addressed by the BRFSS. As a result, some states operate local versions of the BRFSS. An example is the California Health Interview Survey (



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CHIS), which provides information on the health and demographic characteristics of California residents who reside in geographic subdivisions of the state. From time to time, CHIS adds special topics that are of interest to Californians. CHIS is housed at the UCLA Center for Health Policy Research at the University of California, Los Angeles. Data from the BRFSS and CHIS may be downloaded from their respective websites.



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Case Registries

Case Registries

A **registry** is a centralized database for collection of information about a disease. Registries, maintained for many types of conditions, including cancer, are used to track patients and to select cases for case-control studies.

An example of a cancer registry is the National Program of Cancer Registries (NPCR), which is administered by the CDC. This program "... collects data on the occurrence of cancer; the type, extent, and location of the cancer; and the type of initial treatment."¹⁴ The NPCR covers 96% of the U.S. population through its support of cancer registries in 45 states, the District of Columbia, and U.S. territories and jurisdictions.

A second example of a registry is the Surveillance, Epidemiology, and End Results (SEER) Program. The National Cancer Institute (NCI) operates the SEER program, which comprises an integrated system of registries in strategic locations across the United States. At present, SEER covers about 48% of the U.S. population. Examples of information collected include demographic characteristics of cancer patients, their primary tumor site, cancer markers (for example, estrogen receptors), cancer staging, treatments, and survival.¹⁵ **Figure 4.10** shows the location of SEER program cancer registries. **Figure 4.11** illustrates the sequence of surveillance of data flow in the SEER program.

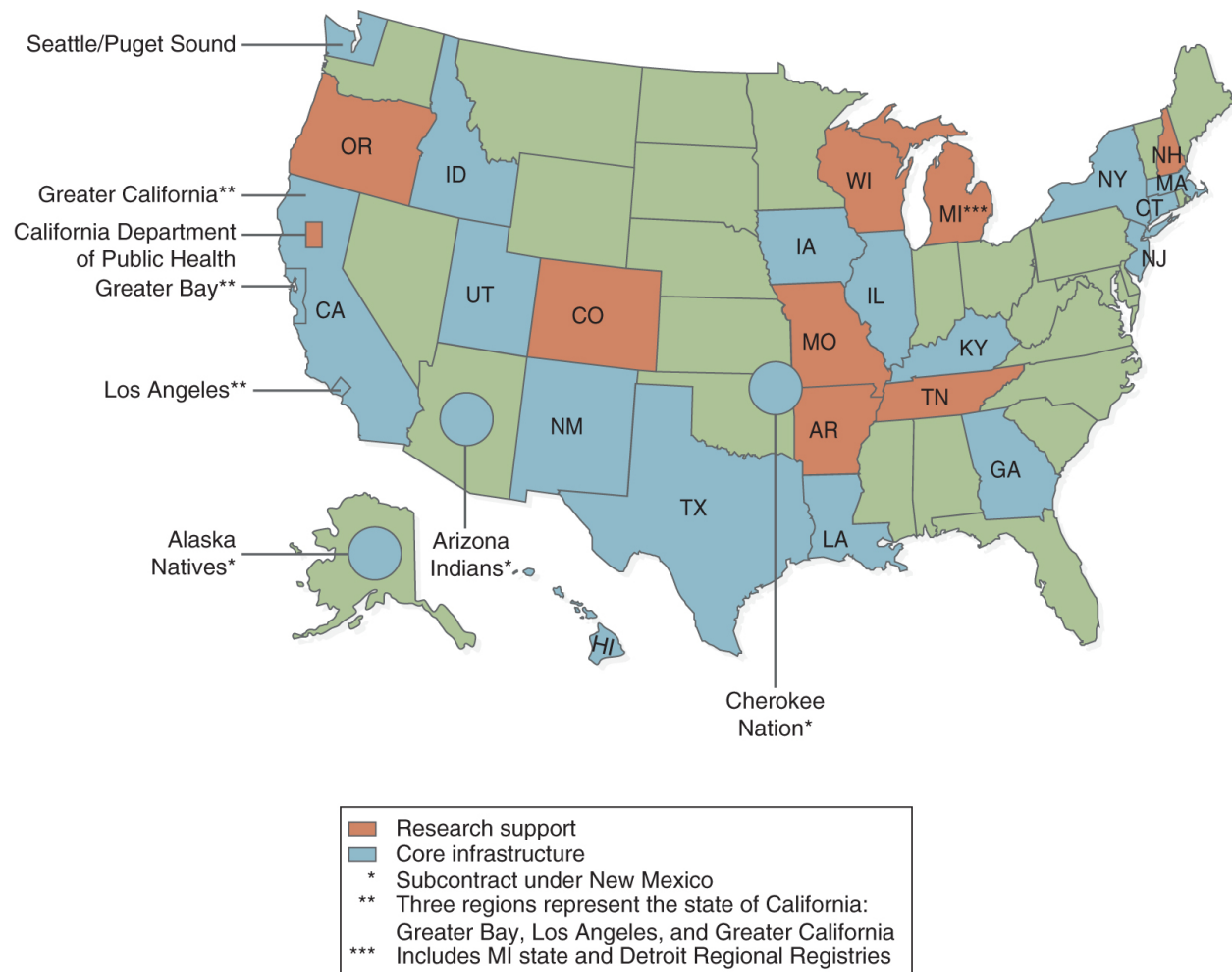


Figure 4.10 SEER program registries

National Cancer Institute. About the SEER registries. Available at: <https://seer.cancer.gov/registries/>.

Accessed February 22, 2023.

Description

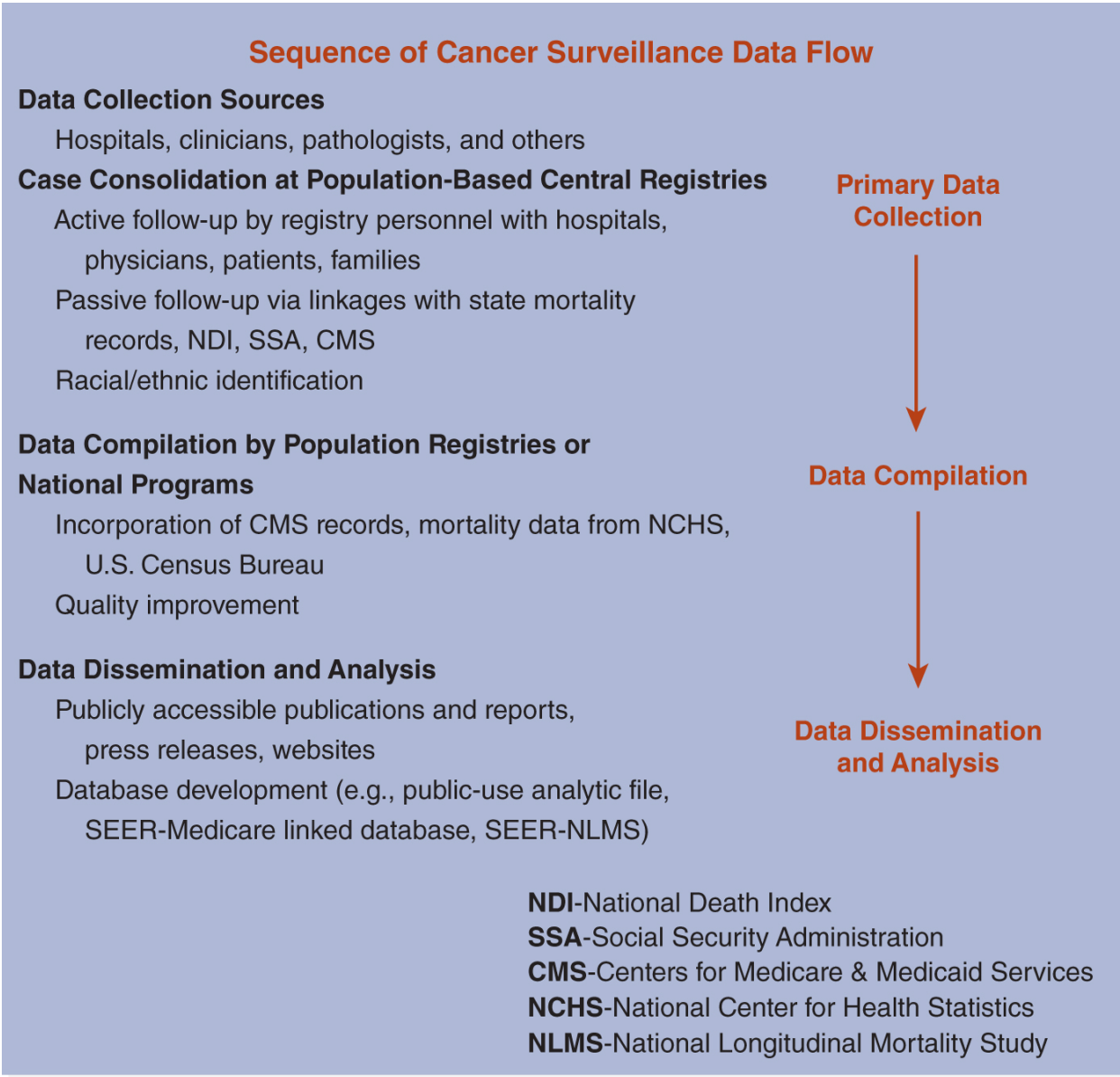


Figure 4.11 The temporal sequence of cancer surveillance data flow.

National Cancer Institute. Data flow in NCI's SEER registries; September 2011. Available at: http://seer.cancer.gov/about/factsheets/SEER_Data_Flow_.pdf. Accessed August 11, 2016.

Description

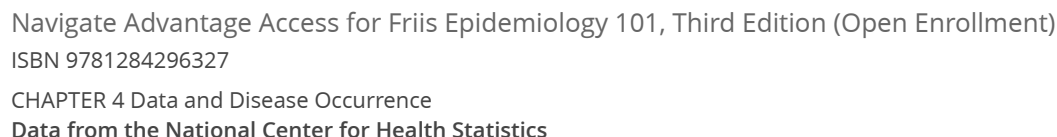
Together, the NPCR and SEER cover the complete U.S. population. Researchers, public health officials, and policy makers regard the data collected by these registries as essential to their work. **Table 4.5** gives an overview of some of the possible uses of data from a cancer registry.

TABLE 4.5 Uses of Data from Cancer Registries

- Monitor cancer trends over time.
- Show cancer patterns in various populations and identify high-risk groups.

- Guide planning and evaluation of cancer control programs.
 - Help set priorities for allocating health resources.
 - Advance clinical, epidemiologic, and health services research.
-

Centers for Disease Control and Prevention. National Program of Cancer Registries (NPCR). Available at: <http://www.cdc.gov/cancer/npcr/about.htm>. Accessed February 27, 2023.



The scope of information available from the NCHS is extensive. Examples of data available from the NCHS through its population surveys are given in **Table 4.6**. Note that both the National Health Interview Survey (NHIS) and the National Health and Nutrition Examination Survey (NHANES) are examples of **morbidity surveys of the population**. The surveys can be designed to elicit information about issues that may not be picked up by other routinely available sources, for example, reportable disease statistics. The NHIS collects information by interview only, while the NHANES also collects information from physical examinations. Such data may disclose undiagnosed conditions not counted by other data collection methods.

	National Health Interview Survey (NHIS)	National Health and Nutrition Examination Survey (NHANES)	National Survey of Family Growth
Data source and methods	Personal interviews	Personal interviews, physical examinations, laboratory tests, nutritional assessment, DNA repository	Personal interviews, men and women 15–44 years of age
Selected data items	■ Health status and limitations ■ Utilization of health care ■ Health insurance ■ Access to care ■ Selected health conditions ■ Poisonings and injuries ■ Health behaviors ■ Functioning/disability ■ Immunizations	■ Selected diseases and conditions, including those undiagnosed or undetected ■ Nutrition monitoring ■ Environmental exposures monitoring ■ Children's growth and development ■ Infectious disease monitoring ■ Overweight and diabetes ■ Hypertension and cholesterol	■ Contraception and sterilization ■ Teenage sexual activity and pregnancy ■ Infertility, adoption, and breastfeeding ■ Marriage, divorce, and cohabitation ■ Fatherhood involvement

Description

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<https://openpage-ebooks.iblearning.com/wr/viewer.html?skipLastRead=true&oneTimePasscode=ST-7e402de2-9644-461e-8d94-35b1848538b1&launchOrgCode=ibl...>

objectives as well as for tracking people's health status and access to health care. The goal of the survey is to collect data from a representative sample of the U.S. population. During each survey wave, interviewers contact up to 40,000 households and obtain data on as many as 100,000 respondents. The interview consists of a set of unvarying core items plus additional questions that change from year to year. Refer again to Table 4.6 for information regarding topics covered by the NHIS. The NCHS releases datasets that contain results from the NHIS and also publishes related documents.



NATIONAL HEALTH INTERVIEW SURVEY— SINCE 1957



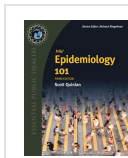
Figure 4.12 The National Health Interview Survey

Photos courtesy of CDC/James Gathany.

Description

National Health and Nutrition Examination Survey

NHANES is also under the administrative purview of the NCHS, which is housed within the CDC.¹⁷ The National Health Survey Act of 1956 provided for the creation of studies to characterize illness and disability



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Data from the National Center for Health Statistics

National Health Interview Survey

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The NCHS conducts the NHIS, which has been in operation since 1957.¹⁶ **Figure 4.12** shows the logo of the NHIS. Data from the NHIS are used for monitoring how well the nation is progressing toward specific health objectives as well as for tracking people's health status and access to health care. The goal of the survey is to collect data from a representative sample of the U.S. population. During each survey wave, interviewers contact up to 40,000 households and obtain data on as many as 100,000 respondents. The interview consists of a set of unvarying core items plus additional questions that change from year to year. Refer again to Table 4.6 for information regarding topics covered by the NHIS. The NCHS releases datasets that contain results from the NHIS and also publishes related documents.



NATIONAL HEALTH INTERVIEW SURVEY— SINCE 1957



Figure 4.12 The National Health Interview Survey

Photos courtesy of CDC/James Gathany.

Description

National Health and Nutrition Examination Survey

NHANES is also under the administrative purview of the NCHS, which is housed within the CDC.¹⁷ The National Health Survey Act of 1956 provided for the creation of studies to characterize illness and disability in the United States. NHANES evolved from this act. The special feature of NHANES is the collection of information from physical examinations coupled with interviews. A mobile examination center is used for performing physical examinations, laboratory tests, dental exams, and interviews. The latter may also be conducted in the respondent's home. Starting in 1999, NHANES has operated as a continuous survey with data released in 2-year cycles. The interviews and examinations performed during the 2-year cycle of 2021–2022 each involved approximately 5,000 individuals. Table 4.6 gives additional information regarding NHANES.

National Vital Statistics System (NVSS)

As noted previously, vital events include births, deaths, marriages, divorces, and fetal deaths. The NVSS is employed by the NCHS to collect and distribute information about vital events in the United States.¹⁸ The NCHS contracts with jurisdictions that are responsible for registering vital events and has promoted the development of standardized forms such as death certificates used for reporting vital events. The NCHS relies on information from the NVSS to publish *Vital Statistics of the United States* and *National Vital Statistics Reports* (NVSR). An example of an NVSR publication is *Births: Final Data for 2021*.¹⁹ **Figure 4.13** reports data on changes in preterm birth rates in the United States—information obtained from the NVSS and printed in *Births: Final Data for 2021*. NVSS-related programs include the Linked Birth and Infant Death Data Set, the National Survey of Family Growth, and the National Death Index.

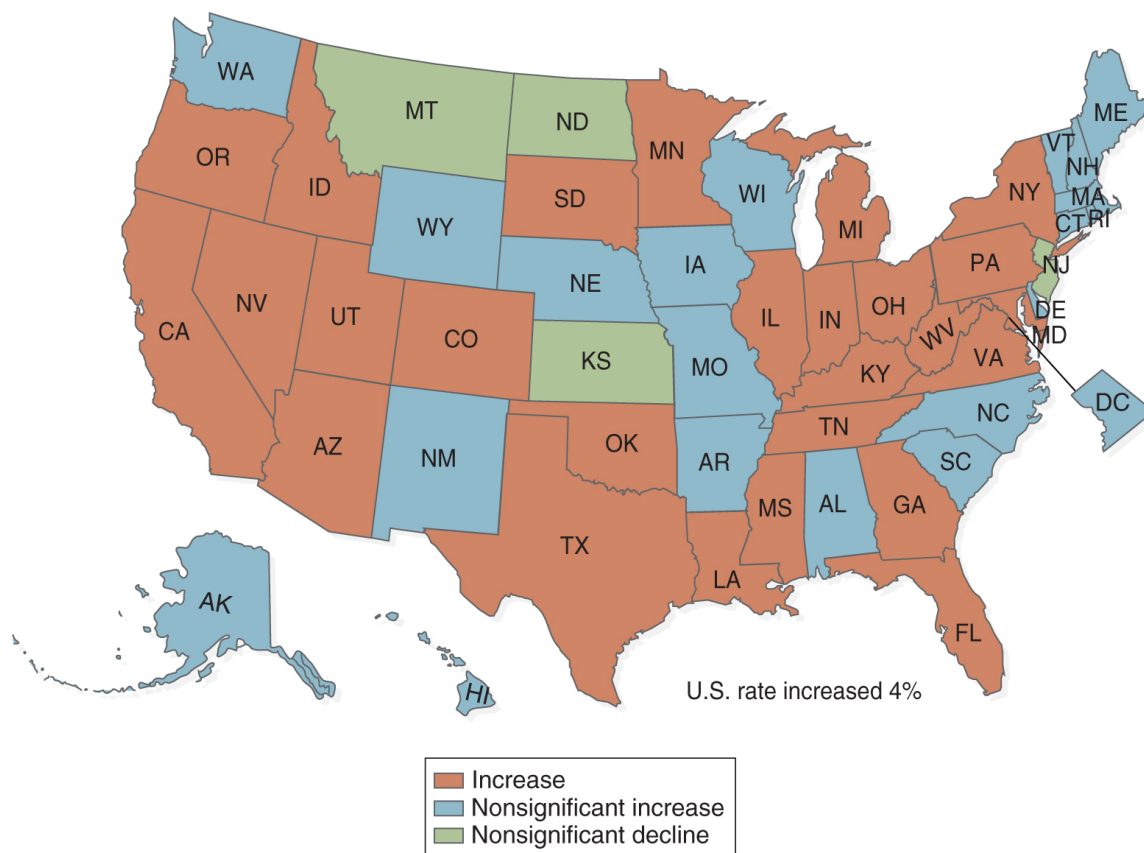


Figure 4.13 Changes in preterm birth rates, by state, 2020 and 2021

Osterman MJK, Hamilton BE, Martin JA, Driscoll AK, Valenzuela CP. *Births: Final data for 2021*.

National Vital Statistics Reports; vol 72, no 1. Hyattsville, MD: National Center for Health Statistics; 2023.

Description